

Quiz 2: Powders 101 — The Workflow and the Chemistry

Review of Session 2 (a merge of the former Sessions 2+3). No new material. ~20–25 minutes.

Section A: The battery

1. List the eight powder tests in the order you run them.
2. Why does running the tests in a fixed order make you faster?
3. How much sample should you use per test?

Section B: Reading results

1. Two powders both fizz with acid. One dissolves fully, one stays cloudy. Name each (with formula).
2. A powder makes an "oobleck" — liquid when stirred slowly, solid when poked fast. What is it, and what will iodine do to it?
3. A powder forms cubic crystals, dissolves fully, doesn't fizz, no iodine reaction, and crackles without melting on gentle heat. What is it?
4. What are the three parts of an ID answer that earn full points?

Section C: Dichotomous keys

1. In a key for {baking soda, chalk, cornstarch, salt, sugar, gypsum}, which single test separates the most powders at once?
2. Trace a key: a powder stays cloudy in water, does not fizz with acid, and turns iodine blue-black. What is it?

Section D: Mixtures

1. Name two signs that a "sample" is actually a mixture of two powders.
2. If a suspected mixture is really one powder, how do you know?

Section E: Formulas

Give the chemical formula:

1. Baking soda
2. Sodium carbonate (washing soda)
3. Calcium carbonate (chalk)
4. Sucrose (table sugar)

5. Starch (cornstarch)
6. Table salt
7. Gypsum
8. Epsom salt
9. Citric acid
10. Vitamin C (ascorbic acid)
11. Sand
12. Sodium acetate

Section F: The iodine test

1. In one or two sentences, explain why starch turns blue-black with iodine.
2. Is the iodine–starch test a chemical reaction? What is the correct term?
3. Amylose turns iodine blue-black. What color does amylopectin (or glycogen) give, and why?
4. What happens to the blue-black color when you heat the mixture? When it cools?
5. Baking soda and baking powder both fizz. One test tells them apart — which, and what result?
6. Vitamin C does something unusual with iodine. What, and why?

Section G: Carbonates and sugars

1. What gas is released when a carbonate reacts with an acid?
2. Write the word equation for baking soda + vinegar.
3. Why does chalk fizz more slowly than baking soda?
4. What does it mean if Benedict's solution turns brick-red?
5. Does table sugar (sucrose) give a positive Benedict's test? Why or why not?

Answer Key

Section A

1. Look, smell, feel, solubility in water, pH, reaction with acid, reaction with iodine, heat.
2. No decisions between tests; you can set up all samples and batch one test across all of them; you don't forget a test.
3. A pinch — about what sticks to a wet toothpick.

Section B

1. Dissolves fully = baking soda, NaHCO_3 . Stays cloudy = calcium carbonate (chalk), CaCO_3 .
2. Cornstarch, $(\text{C}_6\text{H}_{10}\text{O}_5)_n$; iodine turns it blue-black (starch).
3. Table salt (NaCl).
4. Name, chemical formula, and the suspect it implicates (with the reason).

Section C

1. The acid-fizz test — it isolates the two carbonates immediately.
2. Cornstarch (not a carbonate → not step 2; iodine positive → cornstarch).

Section D

1. Two crystal shapes or colors under the lens; a partial test result (some fizz *and* residue; iodine blackens part and leaves grains orange; half-dissolves).
2. There is a single powder that explains *all* of the observations. If no single powder explains everything, it's a mixture.

Section E

1. NaHCO_3 2. Na_2CO_3 3. CaCO_3 4. $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ 5. $(\text{C}_6\text{H}_{10}\text{O}_5)_n$ 6. NaCl
2. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ 8. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ 9. $\text{C}_6\text{H}_8\text{O}_7$ 10. $\text{C}_6\text{H}_8\text{O}_6$ 11. SiO_2 12. $\text{NaC}_2\text{H}_3\text{O}_2$

Section F

1. Amylose coils into a helix; iodine (as triiodide ions) slides inside; the trapped iodine forms a charge-transfer complex that absorbs visible light, so it looks blue-black.
2. No — it's an inclusion complex (a clathrate); the iodine is physically caged in the helix, no bonds change.
3. Red-brown — their branches are too short to hold a long iodine chain.
4. The color disappears (the helix uncoils and the iodine leaves); it comes back on cooling (the helix re-forms).
5. Iodine — baking powder contains cornstarch so it turns blue-black; baking soda alone does not.
6. Iodine goes colorless — vitamin C is a reducing agent and converts the iodine to colorless iodide (this one *is* a real reaction).

Section G

1. Carbon dioxide (CO_2).
2. Sodium bicarbonate + acetic acid → sodium acetate + water + carbon dioxide.

3. Chalk is insoluble, so the acid only reaches the surface of each grain.
4. A reducing sugar is present.
5. No — sucrose is a non-reducing sugar (it has no free aldehyde/ketone group) unless it is first hydrolyzed.